

How to solve K-Means Algorithm Numerical?

Q. Apply K(=2)-Means algorithm over the data below

Where data = { (185, 72), (170, 56), (168, 60), (179, 68), (182, 72), (188, 77) } up to two iterations and show the clusters. Initially choose first two objects as initial centroids.

Solution:

Given, number of clusters to be created (K) = 2 say c1 and c2,
number of iterations = 2 and

The given data points can be represented in tabular form as:

Instance	X	Y
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77

Data Points

also, first two objects as initial centroids:

Centroid for first cluster c1 = (185, 72)

Centroid for second cluster c2 = (170, 56)

Iteration 1: Now calculating similarity by using *Euclidean distance* measure as:

$$d(c1, 3) = \sqrt{(185 - 168)^2 + (72 - 60)^2} = \sqrt{(17)^2 + (12)^2} = \sqrt{289 + 144} = \sqrt{433}$$

$$d(c2, 3) = \sqrt{(170 - 168)^2 + (56 - 60)^2} = \sqrt{(2)^2 + (-4)^2} = \sqrt{4 + 16} = \sqrt{20}$$

Here, $d(c2, 3) < d(c1, 3)$

So, data point 3 belongs to c2.

$$d(c1, 4) = \sqrt{(185 - 179)^2 + (72 - 68)^2} = \sqrt{(6)^2 + (4)^2} = \sqrt{36 + 16} = \sqrt{52}$$

$$d(c2, 4) = \sqrt{(170 - 179)^2 + (56 - 68)^2} = \sqrt{(-9)^2 + (-12)^2} = \sqrt{81 + 144} = \sqrt{225}$$

Here, $d(c1, 4) < d(c2, 4)$

So, data point 4 belongs to c1.

$$d(c1, 5) = \sqrt{(185 - 182)^2 + (72 - 72)^2} = \sqrt{(3)^2 + (0)^2} = \sqrt{9}$$

$$d(c2, 5) = \sqrt{(170 - 182)^2 + (56 - 72)^2} = \sqrt{(-12)^2 + (-16)^2} = \sqrt{144 + 256} = \sqrt{400}$$

Here, $d(c1, 5) < d(c2, 5)$

So, data point 5 belongs to c1.

$$d(c1, 6) = \sqrt{(185 - 188)^2 + (72 - 77)^2} = \sqrt{(-3)^2 + (-5)^2} = \sqrt{9 + 25} = \sqrt{34}$$

$$d(c2, 6) = \sqrt{(170 - 188)^2 + (56 - 77)^2} = \sqrt{(-18)^2 + (-21)^2} = \sqrt{324 + 441} = \sqrt{765}$$

Here, $d(c1, 6) < d(c2, 6)$

So, data point 6 belongs to c1.

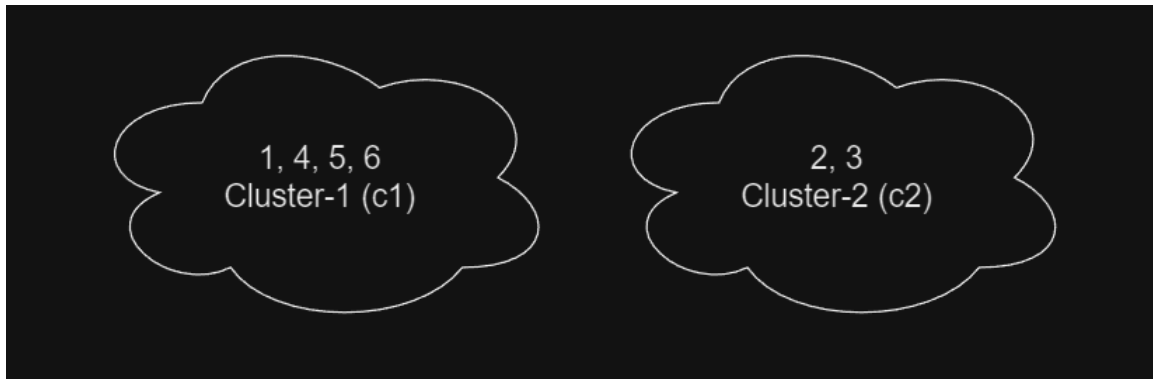
Euclidean distance calculation

Representing above information in tabular form:

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72			c1
2	170	56			c2
3	168	60	$\sqrt{433}$	$\sqrt{20}$	c2
4	179	68	$\sqrt{52}$	$\sqrt{225}$	c1
5	182	72	$\sqrt{9}$	$\sqrt{400}$	c1
6	188	77	$\sqrt{34}$	$\sqrt{765}$	c1

Distance of each data points from cluster centroids

The resulting cluster after first iteration is:



Data points cluster

Iteration 2: Now calculating centroid for each cluster:

$$\text{Centroid for first cluster } c1 = \left(\frac{185+179+182+188}{4}, \frac{72+68+72+77}{4} \right) = (183.5, 72.25)$$

$$\text{Centroid for second cluster } c2 = \left(\frac{170+168}{2}, \frac{56+60}{2} \right) = (169, 58)$$

Calculating centroid as mean of data points

Now, again calculating similarity:

$$d(c1, 1) = \sqrt{(183.5 - 185)^2 + (72.25 - 72)^2} = 1.5207$$

$$d(c2, 1) = \sqrt{(169 - 185)^2 + (58 - 72)^2} = 21.2603$$

Here, $d(c1, 1) < d(c2, 1)$

So, data point 1 belongs to c1.

$$d(c1, 2) = \sqrt{(183.5 - 170)^2 + (72.25 - 56)^2} = 21.1261$$

$$d(c2, 2) = \sqrt{(169 - 170)^2 + (58 - 56)^2} = 2.2361$$

Here, $d(c2, 2) < d(c1, 2)$

So, data point 2 belongs to c2.

$$d(c1, 3) = \sqrt{(183.5 - 168)^2 + (72.25 - 60)^2} = 19.7563$$

$$d(c2, 3) = \sqrt{(169 - 168)^2 + (58 - 60)^2} = 2.2361$$

Here, $d(c2, 3) < d(c1, 3)$

So, data point 3 belongs to c2.

$$d(c1, 4) = \sqrt{(183.5 - 179)^2 + (72.25 - 68)^2} = 6.1897$$

$$d(c2, 4) = \sqrt{(169 - 179)^2 + (58 - 68)^2} = 14.1421$$

Here, $d(c1, 4) < d(c2, 4)$

So, data point 4 belongs to c1.

$$d(c1, 5) = \sqrt{(183.5 - 182)^2 + (72.25 - 72)^2} = 1.5207$$

$$d(c2, 5) = \sqrt{(169 - 182)^2 + (58 - 72)^2} = 19.1050$$

Here, $d(c1, 5) < d(c2, 5)$

So, data point 5 belongs to c1.

$$d(c1, 6) = \sqrt{(183.5 - 188)^2 + (72.25 - 77)^2} = 6.5431$$

$$d(c2, 6) = \sqrt{(169 - 188)^2 + (58 - 77)^2} = 26.8701$$

Here, $d(c1, 6) < d(c2, 6)$

So, data point 6 belongs to c1.

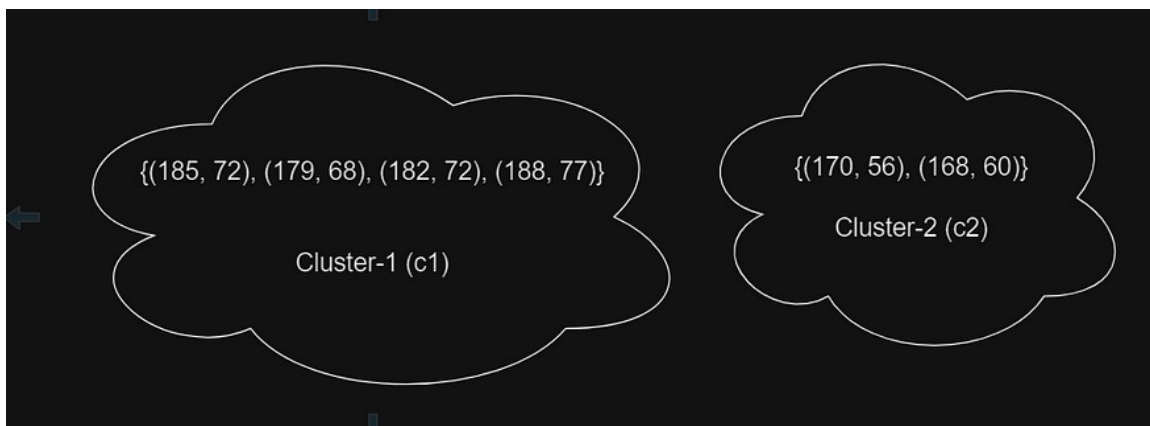
Distance calculation between data points and centroids

Representing above information in tabular form.

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72	1.5207	21.2603	c1
2	170	56	21.1261	2.2361	c2
3	168	60	19.7563	2.2361	c2
4	179	68	6.1897	14.1421	c1
5	182	72	1.5207	19.105	c1
6	188	77	6.5431	26.8701	c1

Distance of each data points from cluster centroids

The resulting cluster after second iteration is:



Data points cluster

As we have already completed two iteration as asked by our question, the numerical ends here.

Since, the clustering doesn't change after second iteration, so terminate the iteration even if question doesn't say so.